Formal Explainability Approach to Diabetes Risk Prediction Using MaxSAT and Wearable-Clinical Data Integration

Caroline Wang

Contents

1 Introduction

Diabetes is a growing public health concern. In 2021, an estimated 1 in 20 Australians (5.1%, or just over 1.3 million people) were living with diagnosed diabetes (Australian Institute of Health and Welfare, 2023). This chronic disease contributes substantially to the national burden of disease, particularly in terms of disability-adjusted life years (DALY), and exerts growing pressure on the healthcare system. In 2020–21, diabetes accounted for approximately $3.4 billion in healthcare expenditure in Australia, representing 2.3% of total disease spending (Australian Institute of Health and Welfare, 2023). Due to its high prevalence and cost, early disease prevention has become a national health priority (Razzak et al., 2019). Despite the growing availability of digital health technologies, many existing diabetes risk prediction systems still rely solely on static clinical data. These models often fail to incorporate critical behavioral factors, such as physical inactivity, poor sleep, and other daily lifestyle patterns, which significantly influence diabetes risk (American Diabetes Association Professional Practice Committee, 2022). Moreover, while recent research and industry applications have started integrating explainable AI (XAI) techniques to improve model transparency, commonly used methods such as SHapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME) (Vilone & Longo, 2021) rely on approximations. Although widely adopted, these tools are heuristic in nature and may not fully address the deeper interpretability challenges posed by complex black-box algorithms. They often lack formal guarantees, which is especially critical in high-stakes domains like healthcare (Rudin, 2019). The lack of formally grounded explanations not only limits clinical adoption but also hinders accountability, trust, and decision-making in healthcare applications This project takes an innovative step by proposing a formally explainable, personalized diabetes prediction system that combines wearable and clinical data. By applying formal reasoning with MaxSAT solvers (Ignatiev et al., 2022), the system enhances prediction accuracy while also delivering reliable, case-specific explanations to improve transparency and trust in AI-driven healthcare.

2 Related Work

2.1 Academic Research on Diabetes Prediction

Existing research has established that diabetes is influenced not only by biological or clinical factors but also by lifestyle and behavioral patterns such as physical inactivity, diet, and sleep patterns (Wu et al., 2014). A variety of machine learning techniques have been proposed for diabetes prediction, including decision trees, support vector machines, and neural networks (Larabi-Marie-Sainte et al., 2019). However, many predictive models in the literature, especially those based on widely used datasets such as the Pima Indians Diabetes Dataset (Sisodia & Sisodia, 2018), rely exclusively on static clinical features (e.g., number of pregnancies, glucose level, blood pressure) and fail to incorporate real-time behavioral data.

2.2 Industry solutions

Some commercial platforms—such as Embla Health (Embla Health, n.d.) and Virta Health (Virta Health, n.d.) have demonstrated the feasibility of integrating wearable sensor data with clinical records to improve diabetes prediction and management. These systems collect continuous physiological and behavioral data to offer personalized feedback and health recommendations. However, they are typically proprietary and lack transparency in their predictive logic, making it difficult for users and clinicians to understand or verify the results. Moreover, few systems clearly link behavioral changes to the guidance provided, which limits their educational value and user engagement.

2.3 Industry solutions

Some commercial platforms—such as Embla Health (Embla Health, n.d.) and Virta Health (Virta Health, n.d.) have demonstrated the feasibility of integrating wearable sensor data with clinical records to improve diabetes prediction and management. These systems collect continuous physiological and behavioral data to offer personalized feedback and health recommendations. However, they are typically proprietary and lack transparency in their predictive logic, making it difficult for users and clinicians to understand or verify the results. Moreover, few systems clearly link behavioral changes to the guidance provided, which limits their educational value and user engagement.

2.4 Industry solutions

Recent studies have explored the use of XAI methods in healthcare (Chaddad et al., 2023). For example, the Mayo Clinic initiative employs SHAP to understand and interpret the impact of different variables on stroke prediction (Kumar, 2023). Similarly, Mount Sinai Health System explores the use of XAI techniques to support machine learning models in predicting conditions such as sepsis in intensive care settings and cancer immunotherapy (Mount Sinai Health System, 2025). Despite their contributions, these methods exhibit two main limitations: (1) reliance on complex machine learning models that create "black-box" environments, making it difficult for clinicians to understand or trust the outputs; and (2) instability in explanation outputs—one single patient may receive different explanations from the same model if retrained on slightly different data. This inconsistency can lead to confusion and reduced confidence among clinicians (Yuvaraj & SriPreethaa, 2017). In contrast to SHAP and LIME, which are heuristic explanation methods, this project adopts a formal XAI approach that uses MaxSAT solvers to compute formally guaranteed, minimal, and logically sound justifications for model predictions by leveraging the logical structure of tree ensemble models (Ignatiev et al., 2022). This project directly addresses current limitations through two core innovations. First, it integrates real-time behavioral data from wearable devices with traditional clinical records, enabling a more comprehensive and dynamic representation of individual diabetes risk. Second, it employs a formal explainability framework based on satisfiability reasoning, providing a more robust and verifiable alternative to heuristic XAI methods. These innovations not only improve the accuracy of health risk predictions but also tackle two major challenges in existing systems—namely, the lack of transparency and the instability of explanations. By enhancing interpretability, decision consistency, and user trust, the system is better suited for both clinical decision support and public-facing health education.

3 Business Model

The proposed project has strong real-world applicability within the healthcare sector. It sits at the intersection of digital health, artificial intelligence, and medical applications of machine learning. By transforming opaque predictive models into interpretable decision-making tools, it adds significant value across the healthcare business domain—supporting clinical workflows, regulatory compliance, and user trust. Primary stakeholders of this project include:

- Patients, who will benefit from precise, personalized, and transparent risk predictions that empower early preventive action.
- Clinicians, who will be supported by interpretable decision-support tools that improve diagnostic confidence and communication.
- Public health authorities, who can leverage the system for large-scale monitoring, early intervention planning, and strategic healthcare resource allocation.

Overall, this project demonstrates the potential of data science to address urgent public health challenges in a practical, trustworthy, and scalable manner.

4 Characterising and Analysing Data

4.1 Data sources and characteristics

Dataset 1: Diabetes Patients Data (Clinical Data)
This dataset originates from the National Institute of Diabetes and Digestive and Kidney Diseases, containing real-world clinical measurements related to diabetes.
Source: Kaggle - National Institute of Diabetes and Digestive and Kidney Diseases

Dataset 2: Population-Level Wearable Device Information (Simulated Wearable Data)
This dataset provides synthetic but realistic data representing population-level health metrics collected via wearable devices. It is designed for evaluating health trends and testing algorithms in wearable health analytics.
Source: Mendeley Data - Simulated Wearable Health Metrics

Dataset 3: 2021 BRFSS Survey Data (Behavioral Health Data)
This is a large-scale public health survey providing behavioral, mental, and social determinants of health relevant to chronic disease risk.
Source: Behavioral Risk Factor Surveillance System (BRFSS), Centers for Disease Control and Prevention (CDC), USA
Cleaned version used in this demonstration:GitHub Repository

Table 1: Data Characteristics (4Vs)

Dataset	Volume	Variety	Velocity	Veracity
Clinical Data	768 patient records with 8 structured clinical features	Numeric clinical indicators only, no free text or time series	Static, one-time collection	Real clinical data, widely validated but limited sample
Wearable Data	10,000 synthetic individuals, each with 3 health-related features over 30 days	Behavioral and physiological metrics across multiple domains	Daily-level simulated stream	Structural format, but synthetic and not validated
Behavioral Health Dataset	438,693 responses collected nation-wide, covering 300 behavioral and demographic variables	Wide range of behavioral indicators: diet, physical/mental health, healthcare access, income	Survey-based, collected annually	Officially collected, but subject to self-reporting bias

- **Platform:** RStudio and Pycharm
- **Processing:** R is used for data preparation, processing(classifier package), and visualization(ggplot2). Python is used exclusively for model explainability, including SHAP value interpretation and MaxSAT-based formal reasoning.
- **Storage:** Dataset 1 and Dataset 2 are stored in .csv format and require no special handling. Dataset 3 (BRFSS) is originally in .XPT format and is imported into R using the haven package. All datasets are currently stored locally due to their manageable size. Cloud-based storage (e.g., AWS S3) will be considered for future scalability.

4.2 Data analysis techniques and statistical methods

1. Random Forest

Role: Serves as a baseline model for comparison and global feature importance assessment.

Rationale: Random Forest is particularly well-suited for the structured tabular data used in this project, which includes both clinical variables and aggregated wearable features. It effectively captures nonlinear interactions—such as how high glucose combined with low physical activity may jointly elevate diabetes risk. Unlike logistic regression, Random Forest makes no assumptions of linearity or feature independence, allowing it to model more realistic patterns of health risk.

Expected Outcome: Identifies and ranks key features (e.g., glucose, BMI, physical activity) by importance, and highlights variables that simpler models may overlook.

2. XGBoost Classification

Role: Acts as the primary classification model, producing high-quality predictions for downstream formal explanation.

Rationale: XGBoost is a high-performance gradient boosting framework based on decision trees. It excels at modeling complex relationships across heterogeneous variables. Its built-in regularization helps prevent overfitting on high-dimensional feature spaces. Most importantly, XGBoost's tree-based architecture is fully compatible with MaxSAT explainability frameworks. It also efficiently handles missing or imputed behavioral data, which are common in real-world wearable health scenarios.

Expected Outcome: Delivers strong predictive performance and generates interpretable tree structures that can be formally analyzed using MaxSAT. These structures enable the extraction of rule-based explanations that mirror real-world diabetes risk pathways.

3. MaxSAT-based Explainability

Role: Used to generate formal, rule-based explanations for tree ensemble models.

Rationale: MaxSAT is used to extract concise, logically minimal rules from individual decision trees derived from XGBoost.MaxSAT is particularly effective for translating black-box tree models into transparent, verifiable logic statements.

Expected Outcome: Produces interpretable, rule-based explanations for individual predictions, enabling end-users to understand and validate AI decisions. In this project, MaxSAT formalizes risk patterns derived from both clinical and behavioral features, enhancing trust and accountability in AI-driven healthcare systems.

5 Demonstration

5.1 Dataset Selection

This demonstration uses a merged dataset combining clinical and behavioral data. The dataset retains the original diabetes outcome labels (`Outcome`)and eight clinical variables.

To simulate wearable-based behavioral signals, six summary features were extracted from the **Mendeley dataset**, including the **mean**, **standard deviation**, and **range** of `step count`, `heart rate`, and `sleep duration`.

A total of 768 individuals were randomly sampled to match the Pima dataset, and their behavioral features were merged row-wise as shown in `merged_data`. Although the two sources are not linked at the individual level, this design supports feasibility testing of multimodal data integration.

In addition, we referenced the cleaned 2021 BRFSS Diabetes Health Indicators dataset as `behavioral_data` to evaluate whether the constructed behavioral features reflect real-world population-level trends.

Table 2: Dataset Summary

Dataset	Observations	Features	Missing Values	Data Types
merged_data	768	14	None	Numeric
behavioral_data	236,378	23	None	Numeric

The final `merged_data` dataset:

- **Clinical Features (8):**
  - **Pregnancies**: Number of prior pregnancies
  - **Glucose**: Plasma glucose concentration
  - **BloodPressure**: Diastolic blood pressure (mm Hg)
  - **SkinThickness**: Triceps skin fold thickness (mm)
  - **Insulin**: Serum insulin (mu U/ml)
  - **BMI**: Body mass index
  - **DiabetesPedigreeFunction**: Diabetes pedigree score
  - **Age**: Age in years
- **Wearable-Derived Behavioral Features (6):**
  - **avg_steps**: Average daily step count over 30 days
  - **std_steps**: Standard deviation of step count
  - **avg_hr**: Average heart rate
  - **hr_range**: Heart rate range (max - min)
  - **avg_sleep**: Average sleep duration (hours)
  - **sleep_std**: Sleep duration variability
- **Target Variable:**
  - **Outcome**: Diabetes diagnosis (1 = Yes, 0 = No)

The final `behavioral_data` dataset:

- **Behavioral Features(6)**
  - **PhysActivity**: Participates in physical activity (1 = Yes, 0 = No)
  - **Smoker**: Smoking status (1 = Smoker, 0 = Non-smoker)
  - **HvyAlcoholConsump**: Heavy alcohol consumption (1 = Yes, 0 = No)
  - **Fruits**: Regular fruit consumption (1 = Yes, 0 = No)
  - **Veggies**: Regular vegetable consumption (1 = Yes, 0 = No)
  - **CholCheck**: Cholesterol checked in past year (1 = Yes, 0 = No)
- **Demographic & Health Condition Features(17)**
  - **Age**: Age of the respondent
  - **Sex**: Gender (1 = Male, 0 = Female)
  - **Education**: Education level (1 to 6 scale)
  - **Income**: Income level (ordinal scale)
  - **GenHlth**: General health rating (1 = Excellent to 5 = Poor)

  - **MentHlth**: Days of poor mental health (0–30)
  - **PhysHlth**: Days of poor physical health (0–30)
  - **DiffWalk**: Difficulty walking (1 = Yes, 0 = No)
  - **HighBP**: Diagnosed high blood pressure (1 = Yes, 0 = No)
  - **HighChol**: High cholesterol (1 = Yes, 0 = No)
  - **Stroke**: History of stroke (1 = Yes, 0 = No)
  - **HeartDiseaseorAttack**: History of heart disease or heart attack (1 = Yes, 0 = No)
  - **BMI**: Body Mass Index
  - **AnyHealthcare**: Has access to healthcare (1 = Yes, 0 = No)
  - **NoDocbcCost**: Could not see doctor due to cost (1 = Yes, 0 = No)
  - **children**: Number of children
  - **employe**: Employment level/status

- **Target Variable**
  - **Diabetes_binary**: Target variable — diabetes diagnosis (1 = Yes, 0 = No)

5.2 Exploratory Analysis

5.2.1 Data Distribution

The distribution is moderately imbalanced, with approximately 35% of the records labeled as diabetic(Figure 1).

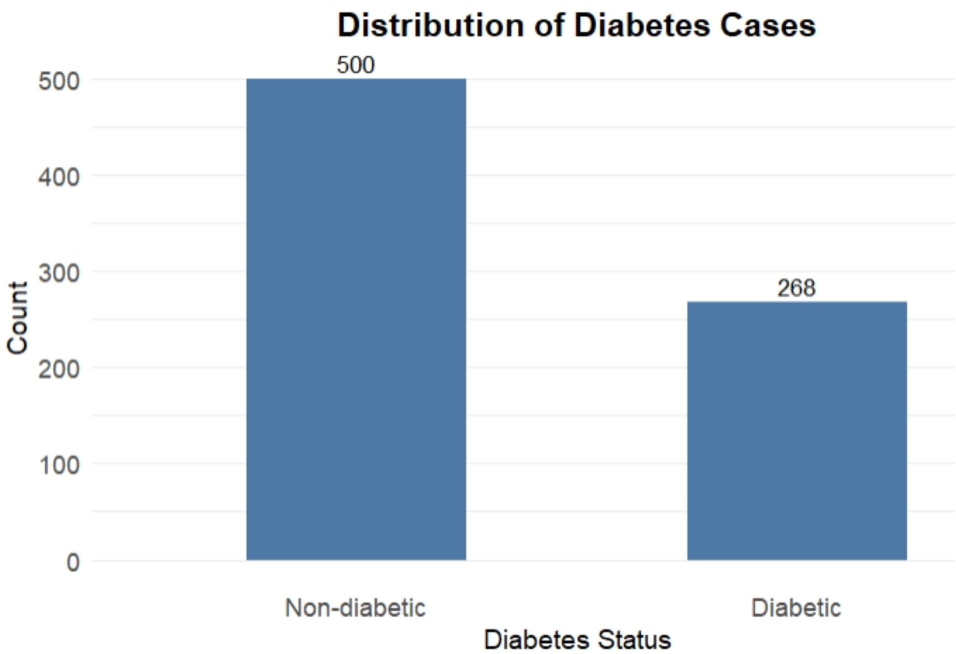

Figure 1: Distribution of Diabetes Cases

Figure 2 shows the density distributions of five key wearable-derived behavioral features. Overall, the distributions appear reasonable, reflecting the variability in physical activity, heart rate, and sleep quality across the sample population.

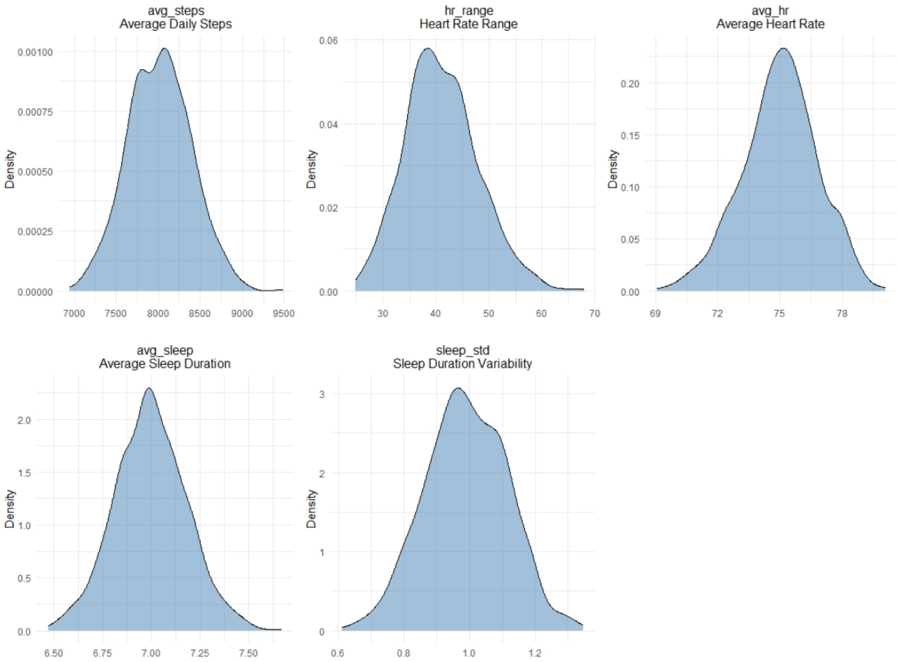

Figure 2: Density plots of key wearable-derived behavioral features

5.2.2 Correlation Heatmaps

A. Merged Dataset Heatmap

To explore the structure of the merged dataset and assess whether the selected features are suitable for modeling, a correlation heatmap was generated.

This heatmap allows us to examine potential multicollinearity, confirm the expected relationships between key clinical variables (e.g., `Glucose`, `BMI`) and the target (`Outcome`), and observe whether the synthetic behavioral features contribute any meaningful variation.

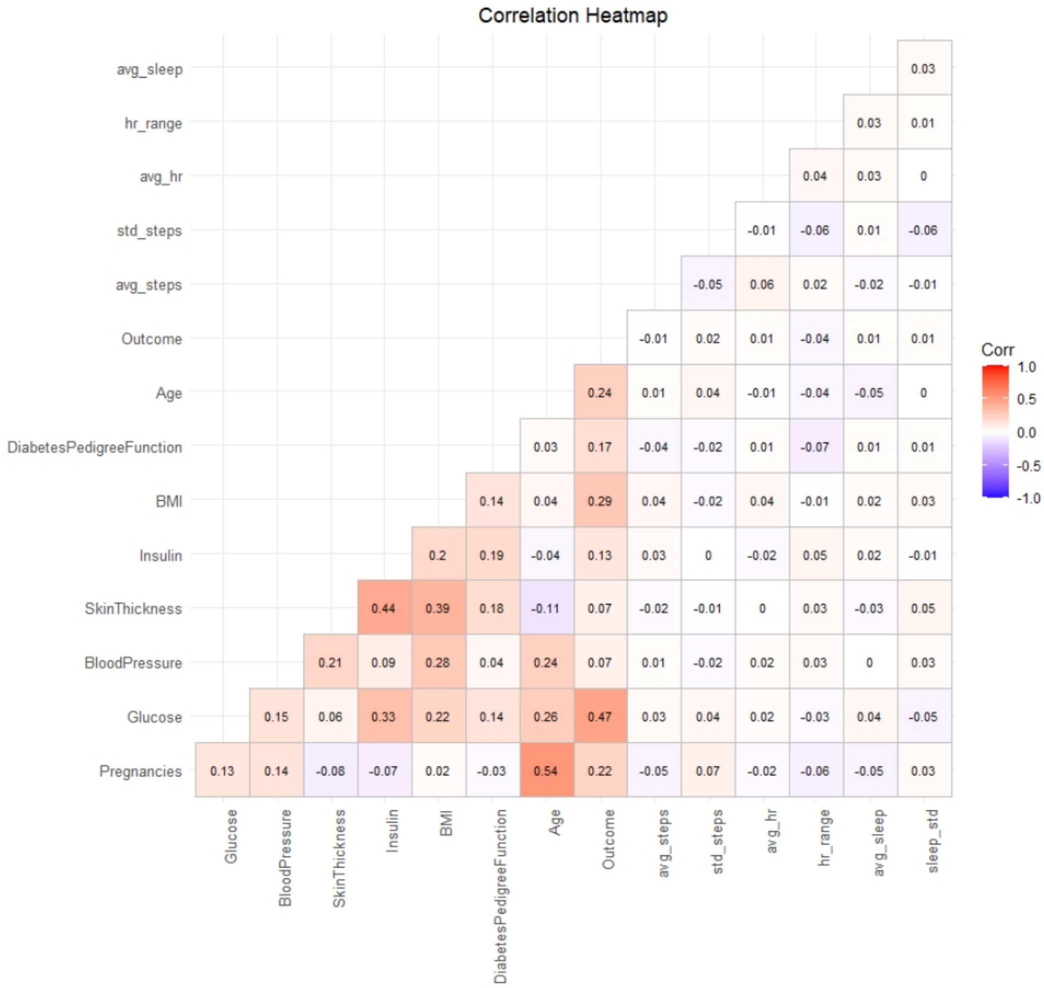

Figure 3: Correlation heatmap of the merged dataset containing clinical and wearable-derived features.

As shown in Figure 3, no pair of features shows a correlation above 0.8, suggesting that multicollinearity is not a concern, and the feature structure is suitable for machine learning modeling.

The correlation heatmap reveals a clear and expected structure among clinical variables.

In contrast, wearable-derived features (including `avg_steps`, `avg_hr`, `avg_sleep`, and `sleep_std`) show little to no linear correlation with the outcome, with most r values ranging from -0.05 to 0.03.

B. BRFSS Behavioral Heatmap

In the previous correlation heatmap, no strong linear associations were observed. To assess the real-world relevance of behavioral features, we generated a correlation heatmap using key variables from the 2021 BRFSS Diabetes dataset. This heatmap examines the associations between behavioral indicators and diabetes status in a large population sample.

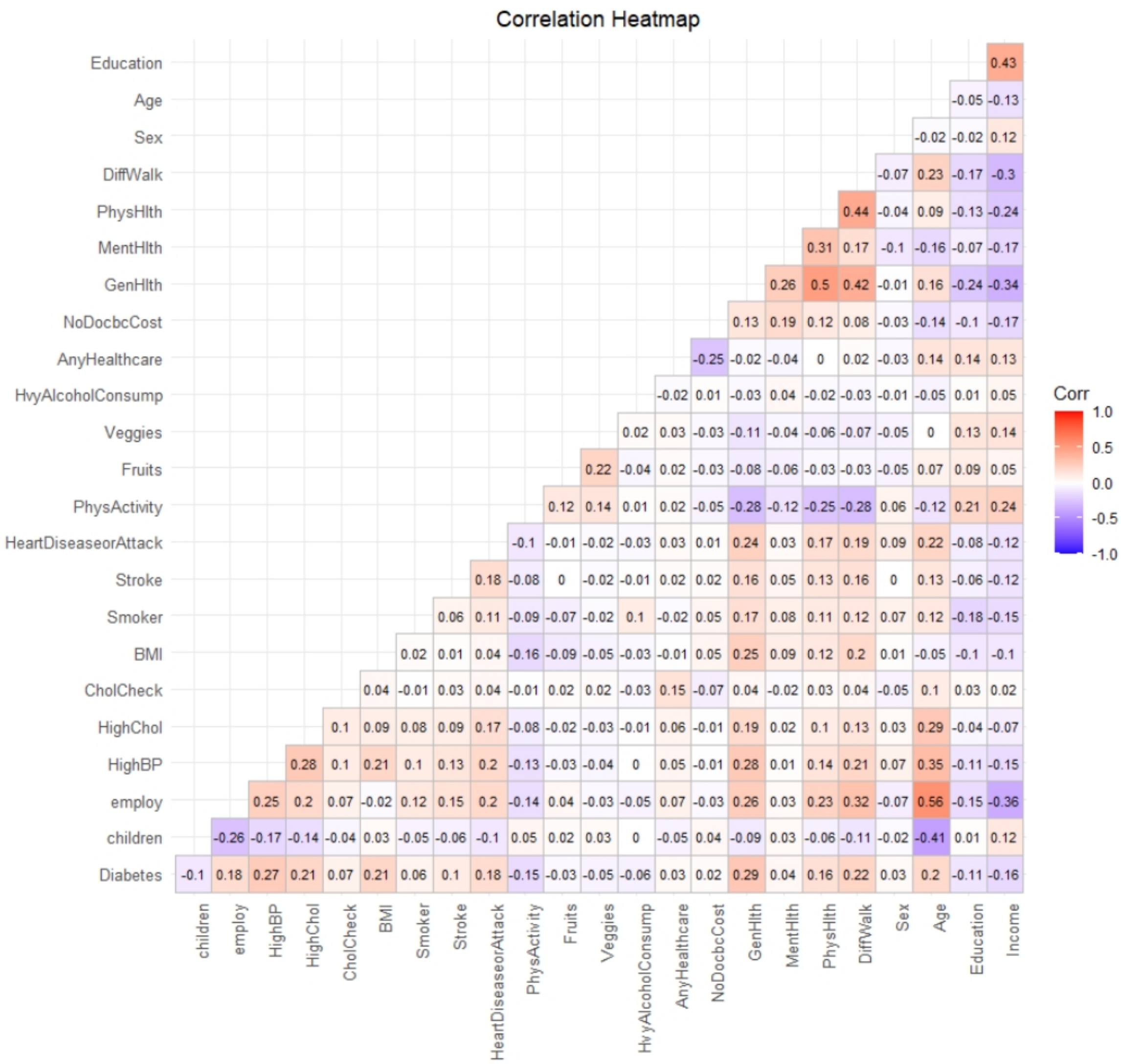

Figure 4: Correlation heatmap of behavioral features and diabetes status in the 2021 BRFSS dataset.

As shown in Figure 4,Among the six behavioral features selected from the BRFSS dataset, `General Health` ($r = 0.26$) and `Difficulty Walking` ($r = 0.23$) show the strongest positive correlations with diabetes status, indicating that poorer health perception and limited mobility are associated with higher diabetes prevalence.In contrast, `Physical Activity` is negatively correlated with diabetes ($r = -0.15$), suggesting a potential protective effect.

These patterns support the relevance of behavior-related features in diabetes risk and justify their inclusion in multimodal predictive models.

5.2.3 Comparative Modeling

A. Random Forest Model Comparison

Two **Random Forest models** were trained and evaluated to assess the impact of wearable-derived features on diabetes prediction.

- **Baseline model using `clinical_data` (8 clinical features)**

- **Merged model using `merged_data` (8 clinical + 6 wearable features)**

Both models were trained on a 70% training set using stratified sampling (createDataPartition()), and evaluated on the same test set.

Results are shown below:

Table 3: Random Forest Model Performance Comparison

Model	**Accuracy**	**AUC**
Clinical Only	0.7696	0.8277
Clinical + Wearable	0.7957	0.8299

The merged model showed a slight improvement in both accuracy and AUC, suggesting a modest predictive gain from wearable features.

B. XGBoost Model Comparison

We adopted **XGBoost** to further optimize performance and enable formal explainability via MaxSAT. he same training-test splits and feature sets were used.

Results are shown below:

Table 4: XGBoost Model Performance Comparison

Model	**Accuracy**	**AUC**	**Sensitivity**	**Specificity**	**F1**
Clinical Only	0.7609	0.8170	0.6133	0.8323	0.6259
Clinical + Wearable	0.7522	0.8005	0.5733	0.8387	0.6014

Interestingly,XGBoost models exhibited slightly lower performance when behavioral features were added to the clinical dataset. Despite expectations that additional behavioral signals might enhance prediction, the merged models showed a minor drop in Accuracy, AUC, and F1-score, particularly in sensitivity.

6 Result

One major limitation of this experiment is that the wearable behavioral data was randomly row-matched to the clinical dataset, without any real individual-level correspondence. This likely weakened the predictive power of the merged model, especially in the case of XGBoost, which is more sensitive to noisy or misaligned features.

However, our prior analysis of the BRFSS 2021 dataset confirmed that behavioral indicators—such as physical activity and general health status—are indeed positively associated with diabetes risk at the population level. This finding supports the conceptual validity of integrating behavioral and clinical data for disease prediction.

Therefore, the present study should not be interpreted as a failure of multimodal integration, but rather as a demonstration of the critical importance of data alignment. Future research should focus on acquiring real-world, identity-linked datasets that combine wearable and clinical data, and on evaluating integrated and explainable models under such conditions.

Additionally, although the MaxSAT-based formal explanation module was not implemented in this phase, the XGBoost model has been successfully trained and provides a structurally interpretable foundation. This lays the groundwork for incorporating MaxSAT in future work, contributing toward the long-term goal of building explainable and logically verifiable AI systems.

7 Standard for Data Science Process, Data Governance and Management

Standard Data Science Process:

- Defining clear objectives: predicting diabetes risk and enhancing model explainability.
- Collecting and merging data properly: using clinical and wearable data.
- Cleaning data and constructing features: preparing and analyzing the dataset.
- Selecting models: applying Random Forest and XGBoost, with plans for explainability analysis.
- Validating results and summarizing findings to ensure scientific rigor and reliability.

Data Governance:

- The **data owners**, responsible for data collection, validation, and publication, include government agencies such as the CDC and research institutions.
- **Data platforms** such as Kaggle, GitHub, and Mendeley serve as repositories and distribution channels but are not the original data owners.
- In this project, as the **sole data scientist**, I take on core data governance responsibilities, including:
  - Ensuring data compliance,
  - Managing data quality,
  - Protecting privacy.

Data Management:

- Data preprocessing
- Feature engineering
- Model training
- Result analysis

These activities are supported by standardized tools and workflows to ensure effective and efficient data utilization.